

Q1. JEE Main 2026 (23 January Shift 1)

A rectangle is formed by the lines $x = 0, y = 0, x = 3$ and $y = 4$. Let the line L be perpendicular to $3x + y + 6 = 0$ and divide the area of the rectangle into two equal parts. Then the distance of the point $(\frac{1}{2}, -5)$ from the line L is equal to :

- (1) $\sqrt{10}$ (2) $2\sqrt{5}$ (3) $2\sqrt{10}$ (4) $3\sqrt{10}$

Q2. JEE Main 2026 (24 January Shift 2)

Let the angles made with the positive x -axis by two straight lines drawn from the point $P(2, 3)$ and meeting the line $x + y = 6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2 . Then the value of $(\theta_1 + \theta_2)$ is:

- (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{6}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{12}$

Q3. JEE Main 2026 (22 January Shift 2)

Among the statements

(S1) : If $A(5, -1)$ and $B(-2, 3)$ are two vertices of a triangle, whose orthocentre is $(0, 0)$, then its third vertex is $(-4, -7)$

and

(S2) : If positive numbers $2a, b, c$ are three consecutive terms of an A.P., then the lines $ax + by + c = 0$ are concurrent at $(2, -2)$,

- (1) both are incorrect (2) only (S2) is correct
(3) only (S1) is correct (4) both are correct

Q4. JEE Main 2026 (23 January Shift 2)

Let $A(1, 2)$ and $C(-3, -6)$ be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x - y = 14$. If $B(\alpha, \beta)$ and $D(\gamma, \delta)$ are the other two vertices, then $|\alpha + \beta + \gamma + \delta|$ is equal to

- (1) 6 (2) 9 (3) 3 (4) 1

Q5. JEE Main 2026 (21 January Shift 1)

Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC , where the points B and C lie on the lines L_1 and L_2 , respectively, is :

- (1) $15\sqrt{6}$ (2) $21\sqrt{3}$ (3) $12\sqrt{2}$ (4) 27

Q6. JEE Main 2026 (28 January Shift 1)

Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x + 2\sqrt{2}y = 4$. If the co-ordinates of the vertex A are (α, β) , then the greatest integer less than or equal to $|\alpha + \sqrt{2}\beta|$ is

- (1) 2 (2) 5 (3) 3 (4) 4

Q7. JEE Main 2026 (24 January Shift 1)

Let $A(1, 0)$, $B(2, -1)$ and $C\left(\frac{7}{3}, \frac{4}{3}\right)$ be three points. If the equation of the bisector of the angle ABC is $\alpha x + \beta y = 5$, then the value of $\alpha^2 + \beta^2$ is

(1) 13

(2) 10

(3) 5

(4) 8

ANSWERS AND SOLUTIONS

1. (3) 2. (3) 3. (4) 4. (1) 5. (2) 6. (4) 7. (2)

1. (3) Rectangle: $x = 0, y = 0, x = 3, y = 4$, area = 12. Center = $(3/2, 2)$.

Slope of $3x + y + 6 = 0$ is -3 , so slope of $L = 1/3$.

A line through the center of a rectangle bisects its area. So L passes through $(3/2, 2)$ with slope $1/3$:

$$y - 2 = \frac{1}{3}(x - 3/2)$$

$$\Rightarrow x - 3y + \frac{9}{2} = 0$$

$$\Rightarrow 2x - 6y + 9 = 0.$$

Distance from $(1/2, -5)$:

$$d = \frac{|2(1/2) - 6(-5) + 9|}{\sqrt{4 + 36}} \\ = \frac{|1 + 30 + 9|}{\sqrt{40}} = \frac{40}{2\sqrt{10}} = 2\sqrt{10}.$$

2. (3) Using parametric form through $P(2, 3)$: point at distance $r = \sqrt{\frac{2}{3}}$ is $(2 + r \cos \theta, 3 + r \sin \theta)$.

Substituting in $x + y = 6$: $r(\cos \theta + \sin \theta) = 1$, so $\cos \theta + \sin \theta = \sqrt{\frac{3}{2}}$.

$$\sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right) = \sqrt{\frac{3}{2}} \Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) = \frac{\sqrt{3}}{2}.$$

$$\theta + \frac{\pi}{4} = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}.$$

$$\theta_1 = \frac{\pi}{12}, \theta_2 = \frac{5\pi}{12}.$$

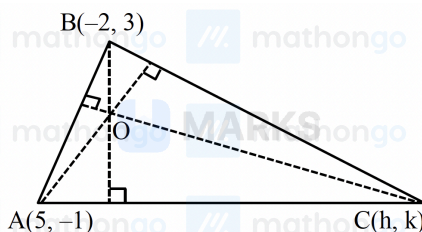
$$\theta_1 + \theta_2 = \frac{\pi}{12} + \frac{5\pi}{12} = \frac{\pi}{2}.$$

3. (4) Statement 1

$$m_{AO} \cdot m_{BC} = -1$$

$$B(-2, 3)$$

$$A(5, -1) \quad C(h, k)$$



$$\Rightarrow 5h - k + 13 = 0 \dots (1)$$

$$\& m_{AB} \cdot m_{OC} = -1$$

$$\Rightarrow 4k = 7h \dots (2)$$

$$\Rightarrow \text{third vertex is } (-4, -7)$$

$\therefore$ Statement 1 is correct.

Statement 2

$$2a, b, c \rightarrow \text{A.P.}$$

$$b = \frac{2a + c}{2}$$

$$\Rightarrow 2a - 2b + c = 0$$

$\therefore$ lines $ax + by + c = 0$ are concurrent then

$$\frac{x}{2} = \frac{y}{-2} = \frac{1}{1}$$

$$x = 2 \text{ and } y = -2$$

$\therefore$ Point of concurrency is $(2, -2)$

$\therefore$ Statement 2 is correct.

4. (1) Midpoint of diagonal AC is $M = (-1, -2)$. Slope of $AC = \frac{-6-2}{-3-1} = 2$.

In a rhombus, diagonals are perpendicular bisectors. So BD has slope $-\frac{1}{2}$ and passes through M :

$$y + 2 = -\frac{1}{2}(x + 1).$$

$$AD \text{ has slope } 7: \frac{\delta - 2}{\gamma - 1} = 7 \Rightarrow \delta = 7\gamma - 5.$$

$$D \text{ lies on } BD: \delta = -\frac{\gamma}{2} - \frac{5}{2}.$$

$$\text{Solving: } 7\gamma - 5 = -\frac{\gamma}{2} - \frac{5}{2} \Rightarrow \gamma = \frac{1}{3}, \delta = -\frac{8}{3}.$$

$$B = (2(-1) - \frac{1}{3}, 2(-2) + \frac{8}{3}) = (-\frac{7}{3}, -\frac{4}{3}).$$

$$|\alpha + \beta + \gamma + \delta| = |-\frac{7}{3} - \frac{4}{3} + \frac{1}{3} - \frac{8}{3}| = |-6| = 6.$$

5. (2) Let $L_2 : y = 0$, $L_1 : y = 9$.

Point A at $(0, 3)$, B at $(x_B, 9)$, C at $(x_C, 0)$.

$$AB^2 = x_B^2 + 36, AC^2 = x_C^2 + 9, BC^2 = (x_B - x_C)^2 + 81.$$

$$\text{From } AB = AC: x_B^2 - x_C^2 = -27.$$

$$\text{From } AB = BC: 2x_Bx_C = x_C^2 + 45.$$

$$\text{Solving: } 3x_C^4 - 198x_C^2 - 2025 = 0.$$

$$\text{Let } u = x_C^2: u^2 - 66u - 675 = 0 \Rightarrow u = 75.$$

$$a^2 = x_C^2 + 9 = 84.$$

$$\text{Area} = \frac{\sqrt{3}}{4} \times 84 = 21\sqrt{3}.$$

6. (4) Let ABC be an equilateral triangle with orthocenter at the origin $O(0,0)$ and side BC on the line $x + 2\sqrt{2}y = 4$.

Since $m_{BC} \cdot m_{AD} = -1$:

$$\left(-\frac{1}{2\sqrt{2}}\right) \left(\frac{\beta}{\alpha}\right) = -1 \Rightarrow \beta = 2\sqrt{2}\alpha \dots (1)$$

$$\text{Distance from O to BC: } OD = \frac{|-4|}{\sqrt{1+8}} = \frac{4}{3}$$

$$\text{For equilateral triangle: } AO = \frac{8}{3}, \text{ so } AD = \frac{4}{3} + \frac{8}{3} = 4$$

$$\frac{|\alpha + 2\sqrt{2}\beta - 4|}{3} = 4 \Rightarrow \alpha = \frac{16}{9} \text{ or } -\frac{8}{9}$$

Since $A(\alpha, \beta)$ and $O(0,0)$ lie on the same side of BC (origin gives $0 + 0 - 4 < 0$), we need $\alpha + 2\sqrt{2}\beta - 4 < 0$.

$$(\alpha, \beta) = \left(-\frac{8}{9}, -\frac{16\sqrt{2}}{9}\right)$$

$$||\alpha + \sqrt{2}\beta|| = \left|\left|-\frac{8-32}{9}\right|\right| = \left|\left|\frac{40}{9}\right|\right| = 4$$

7. (2) For angle bisector of angle ABC where $A(1,0)$, $B(2,-1)$, $C(\frac{7}{3}, \frac{4}{3})$.

Direction BA: $(1-2, 0-(-1)) = (-1, 1)$, unit vector $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

Direction BC: $(\frac{1}{3}, \frac{7}{3})$, magnitude $\frac{5\sqrt{2}}{3}$, unit vector $(\frac{1}{5\sqrt{2}}, \frac{7}{5\sqrt{2}})$

Angle bisector direction (sum of unit vectors): $(\frac{-4}{5\sqrt{2}}, \frac{12}{5\sqrt{2}}) \rightarrow$ direction $(-1, 3)$

Normal to bisector: $(3, 1)$. Bisector passes through $B(2, -1)$:

$$3(x-2) + 1(y+1) = 0 \rightarrow 3x + y - 5 = 0$$

Therefore $\alpha = 3, \beta = 1$, so $\alpha^2 + \beta^2 = 9 + 1 = 10$